

George Kenison

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Academic Interests: Linear and Algebraic Dynamical Systems, Theory of Computation, Symbolic and Algebraic Computation, Program Analysis and Verification.

Employment

Assistant Professor., <i>Department of Computer Science, Maynooth University.</i>	May 2026–
Senior Postdoctoral Fellow in Program Verification, Theory, and Algorithms., <i>Department of Computer Science, KU Leuven.</i>	Feb 2026–May 2026
Research Fellow in Computer Science and Informatics., <i>School of Computer Science and Mathematics, Liverpool John Moores University.</i>	2023–2025
Postdoctoral Researcher in Automated Reasoning and Program Analysis., <i>Institute of Logic and Computation, TU Wien.</i>	2021–2023
Postdoctoral Researcher in Infinite-State Systems and Dynamical Systems., <i>Department of Computer Science, University of Oxford.</i>	2018–2021

Education

PhD in Mathematics, <i>University of Warwick.</i> <i>Thesis: Asymptotics in conjugacy classes for free groups. EPSRC doctoral award studentship.</i>	2013–2017
MMath (Masters of Mathematics), <i>University of Warwick, First Class Hons.</i> <i>Dissertation: Periodic orbits of hyperbolic and quasi-hyperbolic toral automorphisms</i>	2009–2013

Peer-Reviewed Conference Papers

- **2026 (with P. C. Bell, R. Niskanen, I. Potapov, and P. Semukhin).** **On Word Representations and Embeddings in Complex Matrices.** In: *Developments in Language Theory, DLT 2026*, pp. 321–334.
- **2024.** **The Threshold Problem for Hypergeometric Sequences with Quadratic Parameters.** In: *51st International Colloquium on Automata, Languages, and Programming, ICALP 2024*, 145:1–145:20.
- **2024 (with S. Hitarth, L. Kovács, and A. Varonka).** **Linear Loop Synthesis for Quadratic Invariants.** In: *41st International Symposium on Theoretical Aspects of Computer Science, STACS 2024*. Vol. 289, 41:1–41:18.
- **2023 (with L. Kovács and A. Varonka).** **From Polynomial Invariants to Linear Loops.** In: *Proceedings of the 2023 International Symposium on Symbolic and Algebraic Computation, ISSAC 2023*, pp. 398–406.
- **2023 (with K. Nosan, M. Shirmohammadi, and J. Worrell).** **The Membership Problem for Hypergeometric Sequences with Quadratic Parameters.** In: *Proceedings of the 2023 International Symposium on Symbolic and Algebraic Computation, ISSAC 2023*, pp. 407–416.
- **2023 (with J. Nieuwveld, J. Ouaknine, and J. Worrell).** **Positivity Problems for Reversible Linear Recurrence Sequences.** In: *50th International Colloquium on Automata, Languages, and Programming, ICALP 2023*, 130:1–130:17.
- **2022.** **On the Skolem Problem for Reversible Sequences.** In: *International Symposium on Mathematical Foundations of Computer Science, MFCS 2022*, 61:1–61:15.
- **2022 (with D. Amrollahi, E. Bartocci, L. Kovács, M. Moosbrugger, and M. Stankovič).** **Solving Invariant Generation for Unsolvable Loops.** In: *Static Analysis: 29th International Symposium, SAS 2022. Radhia Cousot Award winning paper*, pp. 19–43.
- **2021 (with O. Klurman, E. Lefauchaux, F. Luca, P. Moree, J. Ouaknine, M. A. Whiteland, and J. Worrell).** **On Positivity and Minimality for Second-Order Holonomic Sequences.** In: *International Symposium on Mathematical Foundations of Computer Science, MFCS 2021*, 67:1–67:15.
- **2020 (with R. Lipton, J. Ouaknine, and J. Worrell).** **On the Skolem Problem and Prime Powers.** In: *Proceedings of the 45th International Symposium on Symbolic and Algebraic Computation, ISSAC 2020*, pp. 289–296.

Peer-Reviewed Journal Articles

- **2026 (with R. Ait El Manssour, M. Shirmohammadi, A. Varonka, and J. Worrell).** **Determination Problems for Orbit Closures and Matrix Groups.** In: *Proc. ACM Program. Lang.* 10:POPL.
- **2025 (with R. Ait El Manssour, M. Shirmohammadi, and A. Varonka).** **Simple Linear Loops: Algebraic Invariants and Applications.** In: *Proc. ACM Program. Lang.* 9:POPL, pp. 745–771.
- **2024 (with D. Amrollahi, E. Bartocci, L. Kovács, M. Moosbrugger, and M. Stankovič).** **(Un)Solvable loop analysis.** In: *Formal Methods in System Design SAS 2022*.
- **2019 (with R. Sharp).** **Statistics in conjugacy classes in free groups.** In: *Geometriae Dedicata* 198.1, pp. 57–70.
- **2017 (with R. Sharp).** **Orbit counting in conjugacy classes for free groups acting on trees.** In: *Journal of Topology and Analysis* 9.4, pp. 631–647.

Preprints

- (with O. Klurman, E. Lefauchaux, F. Luca, P. Moree, E. C. Sertöz, J. Ouaknine, A. Whiteland, and J. Worrell). **On the Positivity Problem for second-order holonomic sequences.**
- (with J. Konieczny, F. Luca, A. Scoones, M. Shirmohammadi, and J. Worrell). **On the growth of hypergeometric sequences.** arXiv: [2507.22437 \[math.NT\]](#).

Extended Research Visits

Early Career Researcher, *Banach Center, Institute of Mathematics of the Polish Academy of Sciences*, Simon's May 2026 Semester on Continued Fractions, Fractals, Ergodic theory, and Dynamics.

Visitor, *Faculty of Mathematics, Informatics, and Mechanics (MIMUW). University of Warsaw*, June–July 2026 Programme on Series, Automata, Matrices, Symbolic dynamics, and their Applications.

Teaching Experience

Academic Admissions, *University of Oxford*. 2025–2026

- Undergraduate admissions lead interviewer for computer science and joint schools at Wadham College, Oxford.
- Postgraduate admissions lead interviewer for the Department of Computer Science, Oxford.

Co-lecturer for MSc course on Formal Methods, 2022, 2023
Institute of Logic and Computation, Technische Universität Wien.

Co-lecturer for MSc course Probabilistic Model Checking, 2019–2020
Department of Computer Science, University of Oxford.

Stipendiary Lecturer in Pure Mathematics, *St Peter's College, Oxford*. 2018–2020

- Academic tutor for second year undergraduates. Tutorials in *Linear Algebra*, *Integration*, *Group Theory*, and *Graph Theory*. Duties included feedback, assessment, and writing progression reports.
- Undergraduate admissions interviewer for mathematics and joint schools.

Teaching Associate, *School of Mathematics, University of Bristol*. 2017–2018

- Academic tutor for *Linear Algebra*, *Calculus*, *Metric Spaces*, and *Geometry*.

Teaching Assistant, *Mathematics Institute, University of Warwick*. 2013–2017

- Undergraduate supervisor. Small group teaching across the first year mathematics curriculum.
- Support classes in *Analysis*, *Metric Spaces*, *Experimental Maths*, *Mathematical Computing*, and *Dynamical Systems*.

Departmental Award for Outstanding Teaching, *Mathematics Institute, University of Warwick*.

Student Feedback.

- "Always prepared, always cheerful and always willing to go that extra mile in helping students to understand—a true inspiration!"
- "He was engaging, whilst provoking the students to find their own way to the answers."
- "I was involved in an incident in term one and if it wasn't for his support, both [academic and pastoral], I wouldn't have made it through the term and hence the year."
- "George made me feel comfortable asking questions and... his analysis classes were a highlight of my week."

Selected Seminar Talks

Decision Problems for P-finite Sequences, *Warsaw*. July 2025

The Threshold Problem for Hypergeometric Sequences with Quadratic Parameters, *Liverpool*. June 2024

From Polynomial Invariants to Linear Loops, *IRIF, Paris*. Dec 2023

(Un)Solvable Loop Analysis, *Liverpool*. Oct 2023

Decision Problems for Hypergeometric Sequences, *TU Wien and ISTA*. Feb 2023

On the Skolem Problem and Reversible Sequences, *Chalmers and Gothenburg*. July 2022

On the Skolem Problem and Reversible Sequences, *TU Wien and ISTA*. May 2022

On Positivity and Minimality for Second-Order Holonomic Sequences, *Open University*. Sept 2021

On Positivity and Minimality for Second-Order Holonomic Sequences, *TU Wien and ISTA*. Sept 2021

Skolem's Problem and prime powers, *Bristol*. Dec 2018

Selected Conference and Workshop Talks

Positivity problems and equality tests, *Warsaw*, Jun 2026
Workshop on Series, Automata, Matrices, Symbolic dynamics, and their Applications.

Determination Problems for Orbit Closures and Matrix Groups, *Rennes*, Jan 2026
Symposium on Principles of Programming Languages.

The Membership Problem for Hypergeometric Sequences, *Warsaw*, Jun 2025
Workshop on Series, Automata, Matrices, Symbolic dynamics, and their Applications.

The Threshold Problem for Hypergeometric Sequences with Quadratic Parameters, Vienna, International Conference on Reachability Problems.	Sept 2024
The Threshold Problem for Hypergeometric Sequences with Quadratic Parameters, Tallinn, International Colloquium on Automata, Languages, and Programming.	July 2024
The Membership Problem for Hypergeometric Sequences with Quadratic Parameters, Tromsø, International Symposium on Symbolic and Algebraic Computation.	July 2023
On the Skolem Problem for Reversible Sequences, Kaiserslautern, International Conference on Reachability Problems.	Oct 2022
On the Skolem Problem for Reversible Sequences, Vienna, International Symposium on the Mathematical Foundations of Computer Science.	Aug 2022
On Positivity and Minimality for Second-Order Holonomic Sequences, Tallinn, International Symposium on the Mathematical Foundations of Computer Science.	Aug 2021
On the Skolem Problem and Prime Powers, Kalamata, International Symposium on Symbolic and Algebraic Computation.	July 2020

Widening Participation and Outreach

Mathematics in Education and Industry, Problem Solving Matters: tutor and mentor.	Summer 2017
Further Maths Support Programme, STEP/AEA workshops, enrichment days, Royal Institution master-classes, and problem solving classes.	2012–2019

Academic Service

Program Committee Member.

- RP. *International Conference on Reachability Problems* 2025, 2024
- SCSC. *International Workshop on Satisfiability Checking and Symbolic Computation* 2026

External Reviewer/Sub-Reviewer (Conferences).

- CASC. *Computer Algebra in Scientific Computing* 2022
- CONCUR. *International Conference on Concurrency Theory* 2023
- ICALP. *International Colloquium on Automata, Languages, and Programming* 2026, 2025, 2023, 2020
- ISSAC. *International Symposium on Symbolic and Algebraic Computation* 2026, 2024
- LICS. *Symposium on Logic in Computer Science* 2026, 2025
- MFCS. *International Symposium on Mathematical Foundations of Computer Science* 2023, 2021
- POPL. *Principles of Programming Languages* 2023
- SODA. *Symposium on Discrete Algorithms* 2024
- STACS. *Symposium on Theoretical Aspects of Computer Science* 2025, 2023
- STOC. *Symposium on Theory of Computing* 2025, 2024
- TACAS. *Conference on Tools and Algorithms for the Construction and Analysis of Systems* 2023

External Reviewer/Sub-Reviewer (Journals).

- Ex. Countex. *Examples and Counterexamples*
- FI. *Fundamenta Informaticae*
- Int. J. Math. Comput. Sci. *International Journal of Mathematics and Computer Science*
- J. ACM. *Journal of the Association of Computing Machinery*
- J. Symbolic Comput. *The Journal of Symbolic Computation*
- MCS. *Mathematics in Computer Science*
- MCS. *Journal of Mathematics in Computer Science*
- TOPLAS. *ACM Transactions on Programming Languages and Systems*

Workshop Organiser.

- *Workshop on Loop Invariants and Algebraic Reasoning*, ICALP 2025 Satellite Workshop.
- *Workshop on Reachability, Recurrences, and Loops*, ICALP 2023 Satellite Workshop.

Autobóz, Annual Research Camp on Automata, Logic, and Games.

- **Steering Committee**, 2023–
- **Organiser**, Autobóz 2023 (16–22 July) in partnership with the *Highlights Collaborative Research Week*.
- **Speaker**, Autobóz 2023 (Tutorials on Decision Problems for Linear Recurrence Sequences).

Departmental & University Service

Pay and remuneration committee for sessional teachers, Warwick.	2015–2017
Staff & Graduate Student Liaison Committee (Mathematics), Warwick.	2014–2017